

**Maths2**  
6<sup>th</sup> guided works

**Exercise 1**

1. Study the differentiability of the following functions:

$$f(x) = \begin{cases} \sin x \sin \frac{1}{x}, & \text{if } x \neq 0 \\ 0, & \text{if } x = 0 \end{cases}, \quad g(x) = \begin{cases} x^2 \cos x, & \text{if } x \neq 0 \\ 0, & \text{if } x = 0 \end{cases}$$

2. Determine  $a$  and  $b$  in  $\mathbb{R}$  such that the function  $f$  defined on  $[0, +\infty[$  by

$$f(x) = \begin{cases} \sqrt{x}, & \text{if } 0 \leq x \leq 1 \\ ax^2 + x + 1, & \text{if } x > 1 \end{cases}$$

is differentiable on  $]0, +\infty[$ .

**Exercise 2**

Find  $D$ , the domain of differentiability of  $f$ , and calculate  $f'(x)$  on  $D$  in each of the following cases :

$$(1) \sqrt{6-x-x^2}, (2) \ln(3\sqrt{x}-2), (3) \exp\left(\tan \frac{x}{2}\right), \\ (4) \frac{2\sin x - 1}{3\cos x + 1}, (5) \arcsin(3x^2), (6) \arccos(2x^3)$$

**Exercise 3**

Let  $P$  be the polynomial function defined by

$$P(x) = 3x^4 - 11x^3 + 12x^2 - 4x + 2$$

Show that the derivative  $P'$  of  $P$  has at least one root in the interval  $]0, 1[$ .

**Exercise 4**

**I.** By applying the **Mean Value Theorem**, prove the following:

$$1. \forall x > 0 : \frac{1}{1+x} \leq \ln\left(1 + \frac{1}{x}\right) \leq \frac{1}{x}, \\ 2. \forall x \in \mathbb{R} : e^x \geq 1 + x, \\ 3. \forall x \geq 0 : \frac{x}{1+x^2} \leq \arctan x \leq x.$$

**II.** If we take  $\ln 100 = 4,6052$ , show by using the **Mean Value Theorem** that by writing  $\ln 101 = 4,6151$  we commit an error less than  $10^{-4}$ .

**Exercise 5**

**I.** Using Taylor series expansion around 0 and 1, calculate the following limits:

$$(1) \lim_{x \rightarrow 0} \frac{e^{x^2} - \cos x}{x^2}, (2) \lim_{x \rightarrow 0} \frac{\ln(1+x) - \sin x}{x}, (3) \lim_{x \rightarrow 1} \frac{\cos(x-1) - \sqrt{x}}{x-1}$$

**II.** By using Taylor series expansion, check (**without using a ln key of the calculator**) that

$$(1) \ln(1.001) \approx 0.0009995, (2) \ln(0.999) \approx -0.0010005$$